

EXP 7

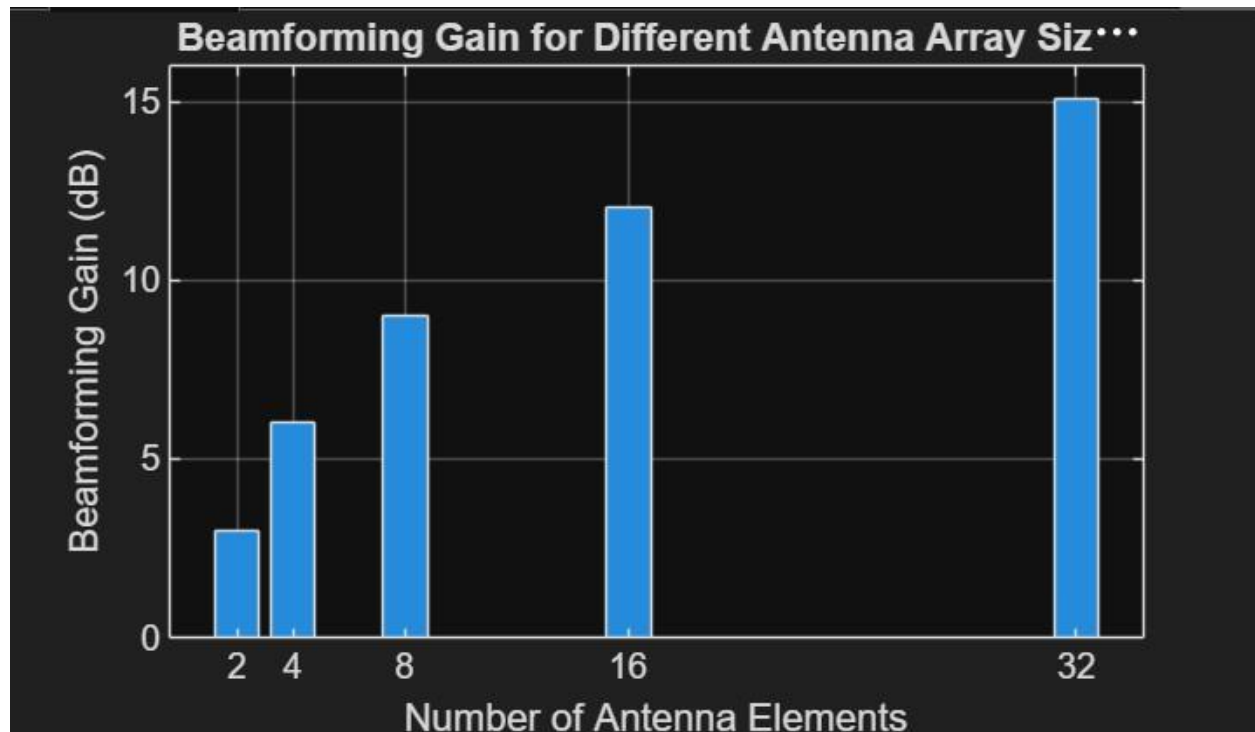
OBJECTIVE 1:

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CODE:

```
clc;
clear;
close all
;
antennas = [2 4 8 16 32]; % Number of Antenna
Elements gain = 10*log10(antennas); % Beamforming
Gain (dB) figure bar(antennas, gain) grid on xlabel(
'Number of Antenna Elements' ) ylabel( 'Beamforming
Gain (dB)' )
title( 'Beamforming Gain for Different Antenna Array Sizes' )
```



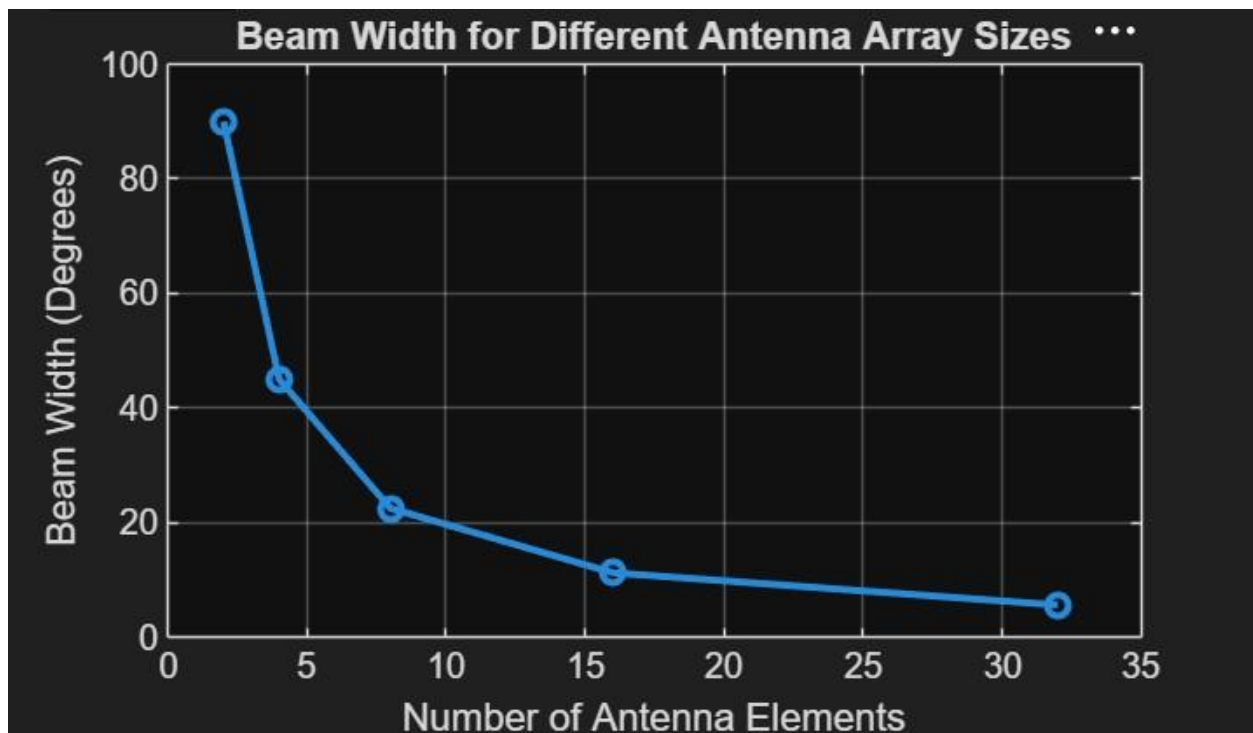
OUTPUT:

OBJECTIVE 2:

CODE:

```
clc;
clear;
close all
;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width
(degrees) figure plot(antennas, beamwidth, 'o-', 'LineWidth', 2)
grid on xlabel( 'Number of Antenna Elements' ) ylabel( 'Beam
Width (Degrees)' )
title( 'Beam Width for Different Antenna Array Sizes' )
```

OUTPUT:



OBJECTIVE 3:

CODE:

```
clc;  
clear;  
close all  
;  
theta = [-60 -30 0 30 60]; % Main Lobe Direction  
(degrees) gain = [1 1 1 1 1]; % Constant normalized gain  
figure polarplot(deg2rad(theta), gain, 'o' , 'LineWidth' ,  
2) title( 'Main Lobe Direction in Analog Beamforming' )
```

OUTPUT:

